

```
!pip install transformers
!pip install sentencepiece
!pip install torch
!pip install pandas
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.3.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.23.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (2025.12.2)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (1.1.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (0.23.0)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.0)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0) (4.12.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (4.10.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (2025.11.3)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (1.0.7)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (3.10.1)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0) (0.14.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (8.2.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (1.3.0)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (13.9.0)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0) (0.0.2)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0) (2.18.0)
Requirement already satisfied: mdurl<=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0) (0.1.2)
Requirement already satisfied: sentencepiece in /usr/local/lib/python3.12/dist-packages (0.2.1)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinjax2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinjax2->torch) (3.0.3)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

```
from transformers import pipeline
import pandas as pd
```

```
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

class CustomTranslator:
    def __init__(self, model_name):
        self.tokenizer = AutoTokenizer.from_pretrained(model_name)
        self.model = AutoModelForSeq2SeqLM.from_pretrained(model_name)

    def __call__(self, text):
        inputs = self.tokenizer(text, return_tensors="pt", padding=True)
        # A default max_length is used, it can be adjusted if needed.
        outputs = self.model.generate(**inputs, max_length=512)
        translated_text = self.tokenizer.decode(outputs[0], skip_special_tokens=True)
        return [{"translation_text": translated_text}]

translator_fr = CustomTranslator("Helsinki-NLP/opus-mt-en-fr")
translator_de = CustomTranslator("Helsinki-NLP/opus-mt-en-de")
translator_hi = CustomTranslator("Helsinki-NLP/opus-mt-en-hi")

print("Models Loaded Successfully")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), s
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json:      1.42k/? [00:00<00:00, 120kB/s]

tokenizer_config.json: 100%                               42.0/42.0 [00:00<00:00, 4.09kB/s]

source.spm: 100%                                          778k/778k [00:00<00:00, 10.5MB/s]

target.spm: 100%                                         802k/802k [00:00<00:00, 41.3MB/s]

vocab.json:      1.34M/? [00:00<00:00, 35.8MB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fa
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
/usr/local/lib/python3.12/dist-packages/transformers/models/arian/tokenization_arian.py:176: UserWarning: Recommended: p
warnings.warn("Recommended: pip install sacremoses.")
pytorch_model.bin: 100%                                  301M/301M [00:02<00:00, 232MB/s]

Loading weights: 100%                                    258/258 [00:00<00:00, 442.67it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                  301M/301M [00:06<00:00, 62.4MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weig
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weig
generation_config.json: 100%                             293/293 [00:00<00:00, 9.52kB/s]

config.json:      1.33k/? [00:00<00:00, 37.6kB/s]

tokenizer_config.json: 100%                               42.0/42.0 [00:00<00:00, 946B/s]

source.spm: 100%                                          768k/768k [00:00<00:00, 6.23MB/s]

target.spm: 100%                                         797k/797k [00:00<00:00, 25.9MB/s]

vocab.json:      1.27M/? [00:00<00:00, 17.4MB/s]

pytorch_model.bin: 100%                                  298M/298M [00:06<00:00, 44.5MB/s]

Loading weights: 100%                                    258/258 [00:00<00:00, 550.26it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                  298M/298M [00:05<00:00, 62.4MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weig
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weig
generation_config.json: 100%                             293/293 [00:00<00:00, 14.9kB/s]

config.json:      1.39k/? [00:00<00:00, 54.9kB/s]

tokenizer_config.json: 100%                               44.0/44.0 [00:00<00:00, 2.28kB/s]

source.spm: 100%                                          812k/812k [00:00<00:00, 11.5MB/s]

target.spm: 100%                                         1.07M/1.07M [00:00<00:00, 43.6MB/s]

vocab.json:      2.10M/? [00:00<00:00, 50.1MB/s]

pytorch_model.bin: 100%                                  306M/306M [00:03<00:00, 160MB/s]

Loading weights: 100%                                    258/258 [00:00<00:00, 476.43it/s, Materializing param=model.shared.weight]

model.safetensors: 100%                                  306M/306M [00:04<00:00, 96.2MB/s]

The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weig
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weig
generation_config.json: 100%                             293/293 [00:00<00:00, 16.4kB/s]
Models Loaded Successfully

```

```

df = pd.read_csv("translation_dataset.csv")
df

```

```

-----
FileNotFoundError                                Traceback (most recent call last)
/tmp/ipykernel_5098/3239367192.py in <cell line: 0>()
----> 1 df = pd.read_csv("translation_dataset.csv")
      2 df

```

```

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/pandas/io/common.py in get_handle(path_or_buf, mode, encoding, compression,
memory_map, is_text, errors, storage_options)
    871     if ioargs.encoding and "b" not in ioargs.mode:
    872         # Encoding
--> 873         handle = open(
    874             handle,
    875             ioargs.mode,

```

**FileNotFoundError:** [Errno 2] No such file or directory: 'translation\_dataset.csv'

```
sentences = df["english_sentence"].tolist()
```

```
print(sentences[:5])
```

```
['The government announced a new education policy today.', 'The minister spoke to the media and explained the new reforms.',
```

```
french_translations = []
```

```

for sentence in sentences:
    translated = translator_fr(sentence)[0]['translation_text']
    french_translations.append(translated)

```

```
df["French"] = french_translations
```

```
df.head()
```

|   | id | category     | sentence_type  | english_sentence                                  | French                                            |
|---|----|--------------|----------------|---------------------------------------------------|---------------------------------------------------|
| 0 | 1  | News         | Simple         | The government announced a new education polic... | Le gouvernement a annoncé aujourd'hui une nouv... |
| 1 | 2  | News         | Compound       | The minister spoke to the media and explained ... | Le ministre s'est entretenu avec les médias et... |
| 2 | 3  | News         | Complex        | Although the economy is growing, unemployment ... | Bien que l'économie se développe, le chômage d... |
| 3 | 4  | Social Media | Conversational | I just watched a great movie and I loved it.      | Je viens de regarder un super film et j'ai adoré. |
| 4 | 5  | Social Media | Interrogative  | Did you complete the assignment for NLP class?    | Avez-vous terminé l'affectation pour la classe... |

```

german_translations = []
hindi_translations = []

```

```

for sentence in sentences:
    german = translator_de(sentence)[0]['translation_text']
    hindi = translator_hi(sentence)[0]['translation_text']

```

```

    german_translations.append(german)
    hindi_translations.append(hindi)

```

```

df["German"] = german_translations
df["Hindi"] = hindi_translations

```

```
df
```

|   | id | category     | sentence_type  | english_sentence                                  | French                                            | German                                            | Hindi                                             |
|---|----|--------------|----------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|---------------------------------------------------|
| 0 | 1  | News         | Simple         | The government announced a new education polic... | Le gouvernement a annoncé aujourd'hui une nouv... | Die Regierung kündigte heute eine neue Bildung... | सरकार ने आज एक नयी शिक्षा नीति घोषित की।          |
| 1 | 2  | News         | Compound       | The minister spoke to the media and explained ... | Le ministre s'est entretenu avec les médias et... | Der Minister sprach mit den Medien und erklärt... | पादरी ने मीडिया से बात की और नए सुधारों को समझ... |
| 2 | 3  | News         | Complex        | Although the economy is growing, unemployment ... | Bien que l'économie se développe, le chômage d... | Obwohl die Wirtschaft wächst, bleibt die Arbei... | हालाँकि बेरोज़गारी बढ़ती जा रही है, फिर भी बेर... |
| 3 | 4  | Social Media | Conversational | I just watched a great movie and I loved it.      | Je viens de regarder un super film et j'ai adoré. | Ich habe mir gerade einen tollen Film angesehe... | मैं सिर्फ एक महान फिल्म देखा और मैं इसे प्यार ... |
| 4 | 5  | Social Media | Interrogative  | Did you complete the assignment for NLP class?    | Avez-vous terminé l'affectation pour la classe... | Hast du den Auftrag für den NLP-Kurs abgeschlo... | क्या आपने एनएलपी क्लास के लिए यह काम पूरा किया?   |
| 5 | 6  | Social Media | Simple         | Learning Python is fun and interesting.           | Apprendre le python est amusant et intéressant.   | Python zu lernen ist lustig und interessant.      | पायथन सीखना मज़ेदार है और दिलचस्प है.             |
| 6 | 7  | Academic     | Technical      | Machine learning models require large amounts ... | Les modèles d'apprentissage automatique nécess... | Maschinelle Lernmodelle erfordern große Datenm... | मशीन सीखने वाले मॉडलों के लिए बहुत मात्रा की आ... |
| 7 | 8  | Academic     | Complex        | When neural networks are trained properly, the... | Lorsque les réseaux neuraux sont correctement ... | Wenn neuronale Netzwerke richtig ausgebildet w... | जब तंत्रिका नेटवर्क को सही तरीके से प्रशिक्षित... |

```
df.to_csv("translated_output.csv", index=False)
```

```
print("File Saved Successfully")
```

```
File Saved Successfully
```

```
9 10 Conversation Simple Please close the door. Fermez la porte, s'il Bitte schließen Sie die Tür.
```

```
from google.colab import files
files.download("translated_output.csv")
```

```
11 12 Conversation Compound I will call you later and we Je vous appellerai plus Ich rufe dich später an und मैं आपको बाद में फोन करूँगे और हम
```

```
complex_sentences = [
    "Although the economy is growing, unemployment remains a concern.",
    "When neural networks are trained properly, they perform better in prediction tasks.",
    "If the dataset is clean, the model will produce accurate results.",
    "Did you complete the assignment for NLP class?",
    "The transformer model uses attention mechanisms for translation."
]
```

```
for sentence in complex_sentences:
    print("English:", sentence)
    print("French:", translator_fr(sentence)[0]['translation_text'])
    print("German:", translator_de(sentence)[0]['translation_text'])
    print("Hindi:", translator_hi(sentence)[0]['translation_text'])
    print()
```

English: Although the economy is growing, unemployment remains a concern.  
 French: Bien que l'économie se développe, le chômage demeure un sujet de préoccupation.  
 German: Obwohl die Wirtschaft wächst, bleibt die Arbeitslosigkeit besorgniserregend.  
 Hindi: हालाँकि बेरोज़गारी बढ़ती जा रही है, फिर भी बेरोज़गारी की चिंता बढ़ती जा रही है।

English: When neural networks are trained properly, they perform better in prediction tasks.  
 French: Lorsque les réseaux neuraux sont correctement formés, ils effectuent mieux dans les tâches de prédiction.  
 German: Wenn neuronale Netzwerke richtig ausgebildet werden, führen sie bessere Vorhersageaufgaben durch.  
 Hindi: जब तंत्रिका नेटवर्क को सही तरीके से प्रशिक्षित किया जाता है, तो वे भविष्यवाणी के कार्यों में बेहतर कार्य करते हैं।

English: If the dataset is clean, the model will produce accurate results.  
 French: Si l'ensemble de données est propre, le modèle produira des résultats précis.  
 German: Wenn der Datensatz sauber ist, liefert das Modell genaue Ergebnisse.  
 Hindi: अगर डेटासेट साफ है, तो मॉडल सही नतीजे पर पहुँचा देगा।

English: Did you complete the assignment for NLP class?  
 French: Avez-vous terminé l'affectation pour la classe NLP?  
 German: Hast du den Auftrag für den NLP-Kurs abgeschlossen?  
 Hindi: क्या आपने एनएलपी क्लास के लिए यह काम पूरा किया?

English: The transformer model uses attention mechanisms for translation.  
 French: Le modèle de transformateur utilise des mécanismes d'attention pour la traduction.